Linear Asymptotics and a Disk-Smearing Bound for the Mordell Distance Product

Fengcan Jiang

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Abstract

For the maximum M_n of the squared distance product under $\sum_{i=1}^n |z_i|^2 = n$, we give an exact finite-n conversion to the ground-state energy of a planar Coulomb gas with quadratic confinement. The deterministic theorem of Sandier and Serfaty then yields $\log M_n = (\frac{1}{2}\log 2 - \frac{1}{4})n^2 + \frac{n}{2}\log n + \gamma n + o(n)$. The microscopic constant satisfies $0.4021788952233 < \gamma < 0.4034264097201$. The lower endpoint is the triangular-lattice comparison of Bétermin and Sandier; the upper endpoint agrees with the classical disk-smearing bound discussed by Lauritsen. A self-contained smearing argument also proves an explicit finite-n upper bound. Its exact deficit splits into Coulomb discrepancy, boundary leakage, and pair-overlap terms. For eight points this gives a concrete positive-gap problem, which remains unsolved.

1 The variational problem and its normalization

All logarithms are natural. For $n \geq 2$, put $N = n(n-1)/2$ and define

$$\Delta_n(z) = \prod_{i<j} |z_i - z_j|^2, \qquad M_n = \max_{\substack{z\in\mathbb{C}^n \\ \sum_i |z_i|^2 = n}} \Delta_n(z). \tag{1.1}$$

Compactness gives a maximizer, and a configuration of distinct points shows that its value is positive. In particular, a maximizer has no collisions. We use the ordered-pair Coulomb Hamiltonian

$$H_n(x) = -\sum_{i\neq j} \log|x_i - x_j| + \frac{n}{2}\sum_i |x_i|^2, \qquad E_n = \min_{\mathbb{C}^n} H_n. \tag{1.2}$$

Thus $H_n = -\log\Delta_n + (n/2)Q$, where $Q = \sum_i |x_i|^2$. The confinement controls the logarithmic interaction at infinity, and collisions have infinite energy, so the minimum is attained away from collisions.

Proposition 1.1 (Exact global conversion)**.** *For every $n \geq 2$,*

$$\log M_n = A_n - E_n, \qquad A_n = N\left[1 - \log\left(1 - \frac{1}{n}\right)\right]. \tag{1.3}$$

Every minimizer of H_n has $Q = n-1$. Multiplication by $\sqrt{n/(n-1)}$ maps such minimizers to maximizers in (1.1)*, and the inverse scaling maps maximizers to minimizers.*

Proof. For every $Q > 0$, homogeneity gives

$$\max_{\sum_i |x_i|^2 = Q} \Delta_n(x) = (Q/n)^N M_n.$$

Consequently E_n is the minimum, over $Q > 0$, of

$$-\log M_n - N\log(Q/n) + nQ/2.$$

Its derivative is $-N/Q + n/2$, and its second derivative is $N/Q^2 > 0$. Its unique minimum is $Q = 2N/n = n-1$. Evaluation proves (1.3) and the assertions about equality and scaling. □

This is a global identity; no prescribed shape or stationary-point classification enters it. Its normalization correction has the convergent expansion

$$A_n - \frac{n^2}{2} = -\frac{1}{2}\sum_{k=0}^{\infty} \frac{n^{-k}}{(k+1)(k+2)} = -\frac{1}{4} - \frac{1}{12n} + O(n^{-2}). \tag{1.4}$$

Keeping the first tail term and bounding the rest geometrically yields

$$-\frac{1}{4} - \frac{1}{12(n-1)} \;\le\; A_n - \frac{n^2}{2} \;\le -\frac{1}{4} - \frac{1}{12n}.$$

In particular, the conversion contributes no additional term of order n. The constant $-1/4$ in this conversion is not an established constant term in the asymptotic expansion of $\log M_n$.

2 The Coulomb theorem and the coefficient gamma

For $g(x) = -\log|x|$, write

$$\mathcal{D}(\mu) = \iint g(x-y)\, d\mu(x)\, d\mu(y), \qquad I_V(\mu) = \mathcal{D}(\mu) + \int V\, d\mu, \quad V(x) = |x|^2/2.$$

There is no factor $1/2$ in $\mathcal{D}$. The equilibrium probability measure and its value are

$$\mu_0 = \frac{1}{2\pi}\mathbf{1}_{\{|x|\le\sqrt{2}\}}\, dx, \qquad I_0 = I_V(\mu_0) = \frac{3}{4} - \frac{1}{2}\log 2. \tag{2.1}$$

Indeed, the logarithmic potential of a uniform disk of radius R equals $\log R - \frac{1}{2} + |x|^2/(2R^2)$ inside the disk and $\log|x|$ outside it. For $R = \sqrt{2}$, its negative plus $V/2$ is constant on the disk and larger outside. The disk's second moment is one, giving (2.1).

We specify the energy convention needed for the linear coefficient. Let $\mathcal{A}_1$ be the class in Sandier–Serfaty [1]: its fields satisfy

$$\operatorname{div} E = 2\pi(\nu - 1), \qquad \operatorname{curl} E = 0, \qquad \nu = \sum_{p\in\Lambda} \delta_p,$$

where Λ is discrete and $\sup_{R>1} \nu(B_R)/|B_R| < \infty$. With the cutoffs χ_R of their Definition 1.3, set

$$W(E) = \limsup_{R\to\infty} \frac{1}{|[-R,R]^2|} \lim_{\eta\downarrow 0} \left[\frac{1}{2}\int_{\mathbb{R}^2\setminus\bigcup_{p\in\Lambda} B(p,\eta)} \chi_R |E|^2 + \pi\log\eta \sum_{p\in\Lambda} \chi_R(p) \right].$$

We write $W_* = \min_{\mathcal{A}_1} W$. The choice of density and of the half-Dirichlet normalization is part of this definition. A different convention changes the numerical value assigned to W_*.

Theorem 2.1 (Linear asymptotic term)**.** *There is a constant, independent of n,*

$$\gamma = -\frac{W_*}{\pi} - \frac{1}{2}\log(2\pi) \tag{2.2}$$

such that

$$\log M_n = \left(\frac{1}{2}\log 2 - \frac{1}{4}\right) n^2 + \frac{n}{2}\log n + \gamma n + o(n). \tag{2.3}$$

Proof. The quadratic potential satisfies the growth and regularity hypotheses (1.11)–(1.13) of [1]: it is smooth, $V/2 - \log|x| \to +\infty$, its Laplacian is the positive constant two, and the equilibrium support has smooth circular boundary. The deterministic conclusion of Theorem 2(C), together with their equations (1.23) and (1.40), therefore gives

$$E_n = n^2 I_0 - \frac{n}{2}\log n + \left[\frac{W_*}{\pi} - \frac{1}{2}\int m_0 \log m_0\, dx\right] n + o(n).$$

Here $m_0 = 1/(2\pi)$ on its support and has total mass one, so the coefficient in brackets is $W_*/\pi + \frac{1}{2}\log(2\pi)$. Substitution into (1.3), using (1.4) and (2.1), proves the claim. □

The proof applies the deterministic ground-state theorem directly. No exchange of the large-n and zero-temperature limits is required. This use of the published theorem, also formulated in Bétermin–Sandier [2], is an external analytic input.

3 A finite-size disk-smearing bound

Let σ_r be uniform probability measure on the disk B_r, and let $T_r f = f * \sigma_r$. Direct angular averaging gives

$$(T_r g)(d) = \begin{cases} -\log r + \frac{1}{2} - \dfrac{d^2}{2r^2}, & 0 \le d < r, \\ -\log d, & d \ge r. \end{cases} \tag{3.1}$$

In particular $T_r g \le g$. Positivity of the averaging operator then gives $T_r^2 g \le T_r g \le g$. These inequalities allow overlapping disks; no point-separation assumption is made. Integration of (3.1) gives the two constants

$$\mathcal{D}(\sigma_r) = -\log r + \frac{1}{4}, \qquad \int |u|^2 \, d\sigma_r(u) = \frac{r^2}{2}. \tag{3.2}$$

Theorem 3.1 (Explicit global upper bound)**.** *For every $n \ge 2$,*

$$M_n \le U_n := n^{n/2} \left(\frac{2n}{n-1} \right)^N e^{-N/2}. \tag{3.3}$$

Consequently

$$\gamma \le \gamma_D := \frac{3}{4} - \frac{1}{2} \log 2. \tag{3.4}$$

Proof. For any collision-free points x_i, set $\mu_x = n^{-1} \sum_i \sigma_r(\cdot - x_i)$. Including and then subtracting exactly n self-terms yields

$$\sum_{i \ne j} g(x_i - x_j) \ge n^2 \mathcal{D}(\mu_x) - n \mathcal{D}(\sigma_r).$$

Moreover, $n^2 \int V \, d\mu_x = (n/2) \sum_i |x_i|^2 + n^2 r^2/4$. Since $I_V(\mu_x) \ge I_0$, (3.2) gives

$$H_n(x) \ge n^2 I_0 + n \log r - \frac{n}{4} - \frac{n^2 r^2}{4}.$$

The right side is maximal at $r^2 = 2/n$. Thus

$$E_n \ge L_n := n^2 I_0 - \frac{n}{2} \log n + n \left(\frac{1}{2} \log 2 - \frac{3}{4} \right). \tag{3.5}$$

The exact conversion (1.3) simplifies to $A_n - L_n = \log U_n$, proving (3.3). Comparison with (2.3) proves (3.4). □

This finite-n proof does not use the asymptotic theorem. For example, (1.4) also gives the strict bound

$$\log M_n < \left(\frac{1}{2} \log 2 - \frac{1}{4} \right) n^2 + \frac{n}{2} \log n + \gamma_D n - \frac{1}{4}.$$

The final $-1/4$ is a correction in this upper bound, not an established boundary correction for the true optimum.

The upper endpoint (3.4) agrees with the classical Lieb–Narnhofer and Sari–Merlini smearing bound discussed in Lauritsen [3]; see Appendix B and Remark 4.5 in the cited arXiv version. We do not claim this universal endpoint as a new constant bound. The preceding argument gives its direct specialization to the finite moment-constrained distance product.

4 The triangular comparison and numerical certificates

Bétermin–Sandier [2, Section 7.3] compute the energy of the unit-density triangular lattice:

$$\frac{1}{\pi}W(\Lambda_\triangle) = \log\pi - \frac{1}{2}\log 3 - 3\log\Gamma(1/3) + \frac{3}{2}\log 2.$$

Because $W_* \le W(\Lambda_\triangle)$, the negative sign in (2.2) gives

$$\gamma_\triangle \le \gamma \le \gamma_D, \qquad \gamma_\triangle = 3\log\Gamma(1/3) + \frac{1}{2}\log 3 - 2\log 2 - \frac{3}{2}\log\pi. \tag{4.1}$$

The companion rational interval certificates give, in particular,

$$\boxed{0.4021788952233 < \gamma < 0.4034264097201.}$$

These decimal endpoints strictly enclose the constants in (4.1); they are not a numerical identification of γ. Identifying γ with $\gamma_\triangle$ would require the triangular value to minimize W over the full admissible class. Optimality within the Bravais-lattice family alone is insufficient.

The program `verify_gamma_bounds.py` certifies the numerical enclosures using rational arithmetic, a Machin formula for π, integer square roots, and controlled exponential, logarithmic and Dedekind-eta series. It evaluates an established analytic lattice formula; it does not prove the Coulomb asymptotic theorem or the global crystallization assertion.

5 An exact nonnegative deficit and the eight-point problem

Keep $r^2 = 2/n$, let $K_r = T_r^2 g$, and define

$$\zeta(\rho) = \begin{cases} 0, & 0 \le \rho \le \sqrt{2}, \\ \rho^2/4 - \log\rho - \frac{1}{2} + \frac{1}{2}\log 2, & \rho > \sqrt{2}. \end{cases} \tag{5.1}$$

This is the equilibrium obstacle excess. It is nonnegative. For $\nu_x = \mu_x - \mu_0$, expansion about μ_0 gives

$$I_V(\mu_x) - I_0 = \mathcal{D}(\nu_x) + 2\int \zeta\, d\mu_x. \tag{5.2}$$

The signed measure ν_x has zero total mass, bounded density and compact support. If $U_{\nu_x} = g * \nu_x$, integration by parts gives

$$\mathcal{D}(\nu_x) = \frac{1}{2\pi}\int_{\mathbb{R}^2} |\nabla U_{\nu_x}|^2\, dx \ge 0.$$

The zero mass eliminates the boundary term at infinity.

Proposition 5.1 (Exact deficit identity)**.** *Every collision-free configuration satisfies*

$$\begin{aligned} H_n(x) - L_n = n^2\mathcal{D}(\nu_x) + 2n^2\int \zeta\, d\mu_x \\ + \sum_{i\ne j}\left[g(x_i - x_j) - K_r(x_i - x_j)\right]. \end{aligned} \tag{5.3}$$

All three terms on the right are nonnegative. The point-pair term vanishes for a pair at distance at least $2r$.

Proof. The exact smoothed-pair sum, including self-terms, is $n^2\mathcal{D}(\mu_x) = \sum_{i\ne j} K_r(x_i - x_j) + n\mathcal{D}(\sigma_r)$. Combine this with the exact smoothed moment formula in the proof of Theorem 3.1, (3.2), and $r^2 = 2/n$. The result is

$$H_n - L_n = n^2[I_V(\mu_x) - I_0] + \sum_{i\ne j}(g - K_r)(x_i - x_j).$$

Now use (5.2). The kernel comparison proves nonnegativity. If $d \geq 2r$, the first smoothed potential agrees with g throughout the second disk, and its second average also equals $g(d)$; hence the pair gap is zero. □

There is also a simple explicit lower estimate. For $d > 0$,

$$g(d) - K_r(d) \geq (1 - d/r)_+^2 + \frac{1}{16}\,(1 - d/(2r))_+^4\,. \tag{5.4}$$

To see this, put $h_r = g - T_r g$. Formula (3.1) gives $h_r(d) \geq (1 - d/r)_+^2$, and $g - K_r = h_r + T_r h_r$. If $d < 2r$, put $\varepsilon = 1 - d/(2r)$. The disk of radius $r\varepsilon/2$ at the midpoint of the two centers lies in their overlap. Its normalized area is $\varepsilon^2/4$, and $h_r \geq \varepsilon^2/4$ there. This proves the second term of (5.4).

For $Q(x) = n - 1$, define $z = \sqrt{n/(n-1)}x$. The conversion (1.3) now holds for this individual configuration as

$$H_n(x) - L_n = \log U_n - \log \Delta_n(z). \tag{5.5}$$

For eight points the conjectured sharp value is $C_8 = 8^{28}/7^{21}$. Thus its global upper bound is equivalent to the following positive-gap assertion over all configurations with $Q(x) = 7$:

$$H_8(x) - L_8 \geq \delta_8, \qquad \delta_8 = \log(U_8/C_8) = 40\log 2 - 7\log 7 - 14. \tag{5.6}$$

Here the smoothing radius is $r = 1/2$. The rational companion certificate establishes, for example,

$$0 < \delta_8 < 0.104517, \qquad 1 < U_8/C_8 < 1.110174.$$

Equation (5.3) therefore reformulates the remaining eight-point problem in terms of three continuous, nonnegative costs. It does not establish the positive gap in (5.6), and does not exclude the general nonzero-moment branch by itself.

6 Verification, attribution and scope

The companion archive contains two standard-library acceptance programs: `verify_gamma_bounds.py` and `verify_finite_bound.py`. They include rational interval certificates. A unified entry point checks source hashes and runs them in a temporary copy:

```
python -B -S verify_all.py.
```

The programs audit constants, algebraic bookkeeping and interval enclosures. The analytic averaging inequalities and the external ground-state theorem remain parts of the mathematical proof; this is not a proof-assistant formalization.

The linear coefficient exists by the published theorem of Sandier–Serfaty, its triangular comparison uses Bétermin–Sandier, and the strong upper endpoint is consistent with the classical smearing literature summarized by Lauritsen. This paper records a normalization audit, a finite-n specialization, and an exact deficit formulation. It makes no global priority claim.

The project and source material are attributed to Fengcan Jiang. Preparation, computation and cross-checking included assistance from OpenAI Codex. AI-assisted checking is not independent external peer review. The exact microscopic minimum, an effective uniform rate in the $o(n)$ remainder, true finite-n boundary corrections, and the unconditional eight-point sharp inequality are not established here.

References

[1] E. Sandier and S. Serfaty, *2D Coulomb Gases and the Renormalized Energy*, arXiv:1201.3503v2 (2013). https://arxiv.org/abs/1201.3503v2.

[2] L. Bétermin and E. Sandier, *Renormalized Energy and Asymptotic Expansion of Optimal Logarithmic Energy on the Sphere*, Constructive Approximation **47** (2018), 39–74; arXiv:1404.4485v5. https://doi.org/10.1007/s00365-016-9357-z.

[3] A. B. Lauritsen, *Floating Wigner crystal and periodic jellium configurations*, Journal of Mathematical Physics **62** (2021), 083305; arXiv:2103.07975v2. https://doi.org/10.1063/5.0053494.